

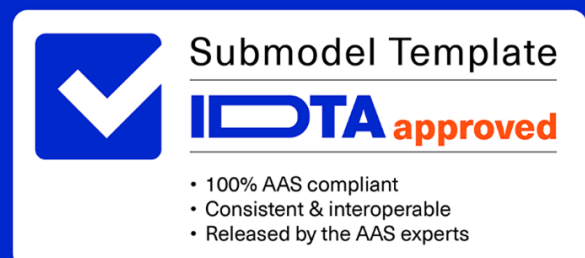
IDTA 02035-5 : Digital Battery Passport - Part 5

Product Condition

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SPECIFICATION

Submodel Template of the
Asset Administration Shell



IDTA 02035-5 V1.0.2

Imprint

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August 2026	1.0.2	Bugfix Release, for details see Changelog in Annex
April 2026	1.0.1	Bugfix Release, for details see Changelog in Annex
18.02.2026	1.0	Release of the official Submodel Template published by IDTA – a joint result of the Model Expert Group comprising IDTA, Catena-X, and the Battery Pass Consortium.

Table of Contents

IDTA 02035-5 V1.0.2	1
Imprint	1
Version history	1
1. General	3
1.1. About this document	3
1.2. Scope of the Submodel	3
1.3. Relevant standards for the Submodel Template	3
1.4. Explanations on used UML diagrams	4
2. Information set for Submodel "ProductCondition"	5
2.1. General	5
2.2. Overview UML model	5
3. Information structures and attributes	8
3.1. Properties of the Submodel "ProductCondition"	8
3.2. Properties of the SMC "EnergyThroughput"	13
3.3. Properties of the SMC "CapacityThroughput"	14
3.4. Properties of the SMC "NumberOfFullCycles"	15
3.5. Properties of the SMC "StateOfCertifiedEnergy"	17
3.6. Properties of the SMC "RemainingEnergy"	18
3.7. Properties of the SMC "RemainingCapacity"	19
3.8. Properties of the SML "NegativeEvents"	20
3.9. Properties of the SML "InformationOnAccidents"	22
3.10. Properties of the SMC "TemperatureInformation"	22
3.11. Properties of the SMC "RemainingPowerCapability"	24
3.12. Properties of the SMC "EvolutionOfSelfDischarge"	26
3.13. Properties of the SMC "CurrentSelfDischargingRate"	27
3.14. Properties of the SMC "RemainingRoundTripEnergyEfficiency"	28
3.15. Properties of the SMC "StateOfCharge"	30
3.16. Properties of the SMC "NegativeEvent"	31
3.17. Properties of the SMC "RemainingPowerCapabilityDynamicAt"	32
Annex A. Explanations on used table formats	34
1. General	34
2. Tables on Submodels and SubmodelElements	34
Annex B. Changes to the submodel template	35
General	35
Changes Version 1.0.2	35
Changes Version 1.0.1	35
Changes Version 1.0	35
Bibliography	36

Chapter 1. General

1.1. About this document

This document is a part of an overall specification series [4]. Each part specifies the contents of a Submodel Template (SMT). The specifications of the Asset Administration Shell (AAS) are the basis for the Submodel Template specifications, see [3].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This specification was created following the "semantic-driven workflow" as defined in [5] based on Aspect Models [6]. There is no central dictionary or repository for Aspect Models. In this specification the following sources are used for defining semantics:

- Aspect Models published at IDTA [7]: <https://github.com/admin-shell-io/smt-semantic-models>, models with namespace "io.admin-shell"
- Aspect Models published at BatteryPass Consortium (closed project) [8]: <https://github.com/batterypass/BatteryPassDataModel>, models with namespace "io.BatteryPass"
- Aspect Models published at Tractus-X and used in standards published by Catena-X [9]: <https://github.com/eclipse-tractusx/sldt-semantic-models>, models with namespace "io.catenax"

1.2. Scope of the Submodel

This Submodel template aims to define the dynamic data points of a battery item required to be included in a Battery Passport conformant to DIN DKE SPEC 99100 and the corresponding EU regulations.

The battery passport consists of the following 7 parts:

Digital Battery Passport - Part 1: Digital Nameplate (IDTA-02035-1)
Digital Battery Passport - Part 2: Handover Documentation (IDTA-02035-2)
Digital Battery Passport - Part 3: Product Carbon Footprint (IDTA-02035-3)
Digital Battery Passport - Part 4: Technical Data (IDTA-02035-4)
Digital Battery Passport - Part 5: Product Condition (IDTA-02035-5)
Digital Battery Passport - Part 6: Material Composition (IDTA-02035-6)
Digital Battery Passport - Part 7: Circularity (IDTA-02035-7)

This specification is Part 5: Product Condition (IDTA_02035-5).

1.3. Relevant standards for the Submodel Template

This submodel template fulfills the requirements for dynamic data points of a battery item as defined in DIN DKE SPEC 99100 [1]. DIN DKE SPEC 99100 "is based on the European Union and key Member States current regulatory requirements for battery passport information. Mandatory information for the battery passport as stated in the EU Battery Regulation (EU)2023/1542, Article77 and AnnexXIII, as well as the Ecodesign for Sustainable Products Regulation (ESPR), is supplemented by recommendations to increase sustainability and circularity. [1]"

This document is valid for all battery categories. Please be aware that for battery categories that have stronger requirements like industrial batteries with battery management systems etc. some of the data points are specified as optional although mandatory per regulation.

1.4. Explanations on used UML diagrams

For clarity and an improved legibility readers suggested to go through this section at first before reading the following chapters.

UML diagrams feature box-like elements, called "classes". These classes, typically Submodels, SubmodelElementCollections or SubmodelElementLists, typically feature a set of Properties or further SubmodelElements. These elements can have specific cardinalities.

The single classes are hierarchally organized by aggregation relations, these can be seen as "contains" relation.

For a further overview on UML diagrams please refer to [2] and [3].

Further details about used table formats please refer to Annex A Explanations.

Chapter 2. Information set for Submodel “ProductCondition”

2.1. General

The "Product Condition" Submodel Template is part of the specification series for the Battery Passport.

Property specification

See clause 3 "Information structures and attributes".

2.2. Overview UML model

The SubmodelElements described in [clause 3 "Information structures and attributes"](#) are structured in the following way (see [Figure 1](#)):

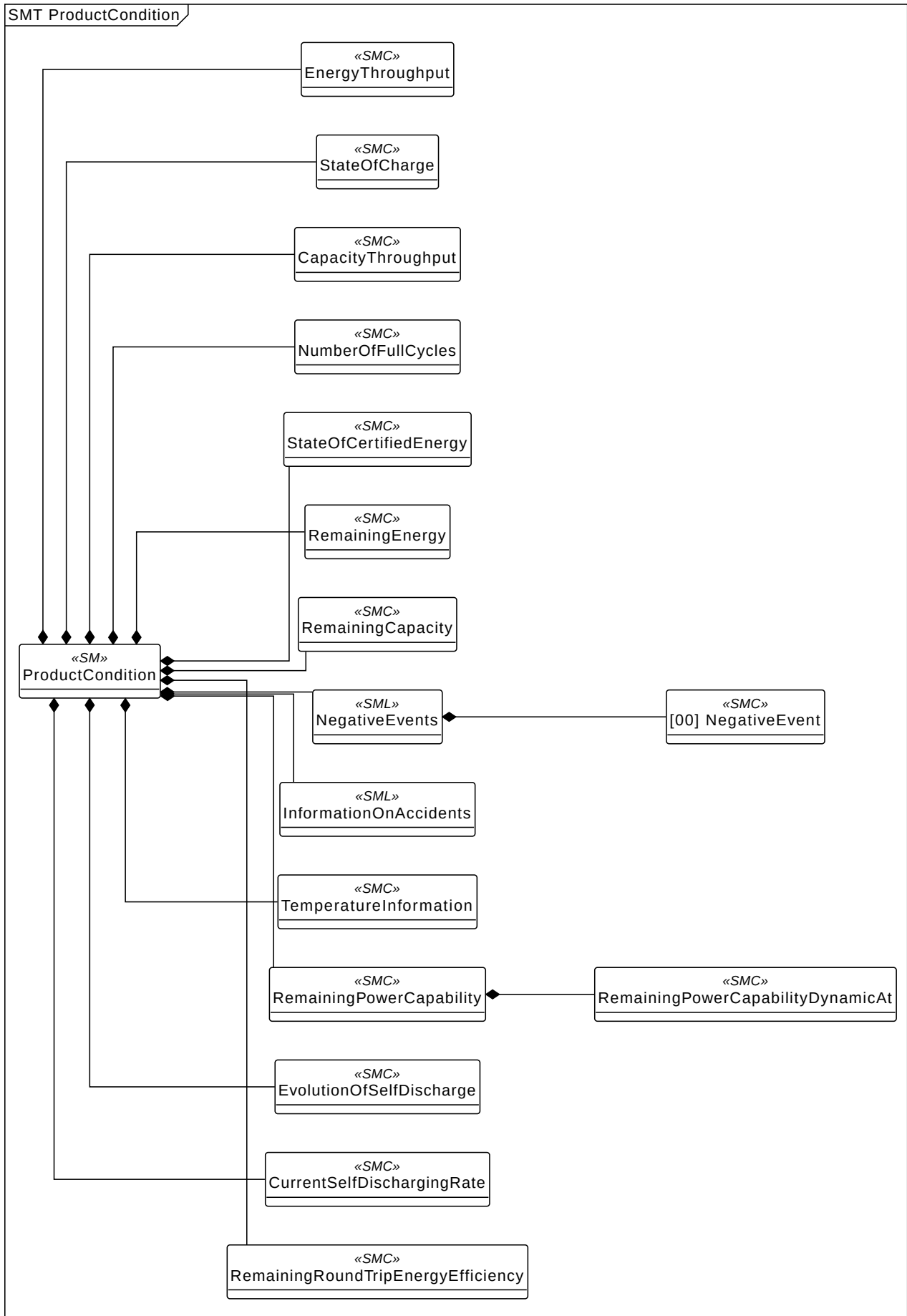


Figure 1. UML overview

The following design principles were followed:

- for all dynamic data points there is an attribute "lastUpdate" showing the last update time

Chapter 3. Information structures and attributes

3.1. Properties of the Submodel “ProductCondition”

Figure 2 shows the UML-diagram defining the relevant properties which need to be set.

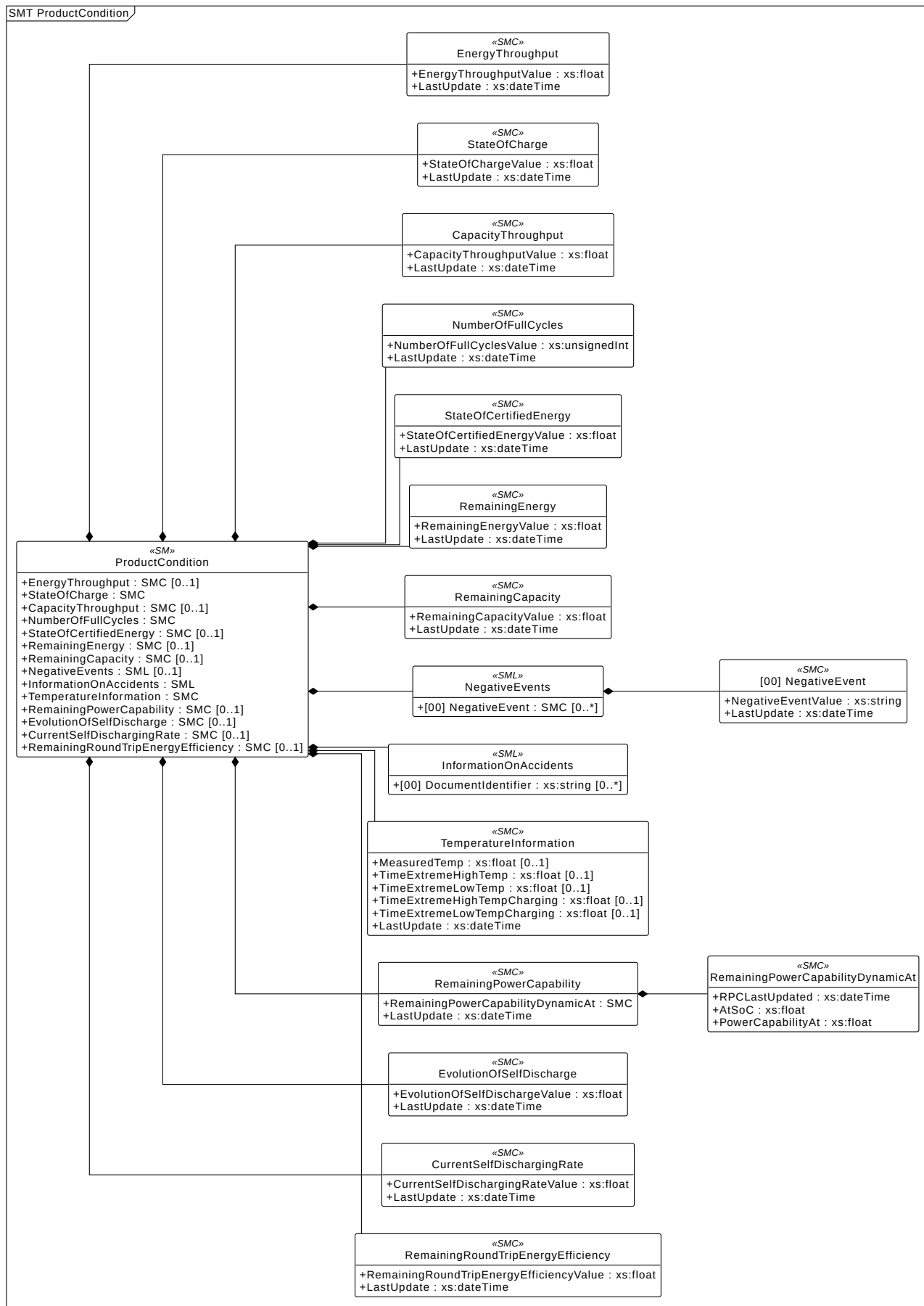


Figure 2. UML-Diagram for Submodel "ProductCondition"

idShort:	ProductCondition
Class:	Submodel

Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#ProductCondition		
Parent:	-		
Explanation:	Covers all battery lifetime relevant properties.		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[SMC] EnergyThroughput	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#energyThroughput supplementalSemanticId(s): urn:irdi:0173-1#02-ABL850#001 The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.7		0..1
[SMC] StateOfCharge	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfCharge isCaseOf: urn:irdi:0173-1#02-ABL821#002 Available capacity in a battery expressed as a percentage of remaining capacity.		1
[SMC] CapacityThroughput	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#capacityThroughput supplementalSemanticId(s): urn:irdi:0173-1#02-ABL849#001 The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.8		0..1

[SMC] NumberOfFull Cycles	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#numberOfFullCycles supplementalSemanticId(s): urn:irdi:0173-1#02-ABL868#001 Number of (full) charging and discharging cycles. DIN DKE Spec 99100 chapter reference: 6.7.6.4		1
[SMC] StateOfCertifiedEnergy	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfCertifiedEnergy supplementalSemanticId(s): urn:irdi:0173-1#02-ABL847#001 Definition based on UNECE GTR 22: The measured or on-board UBE performance at a specific point in its lifetime, expressed as a percentage of the certified usable battery energy. DIN DKE Spec 99100 chapter reference: 6.7.2.7		0..1
[SMC] RemainingEnergy	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingEnergy supplementalSemanticId(s): urn:irdi:0173-1#02-ABL870#001 Definition from UNECE GTR 22, applicable only to EVs. The energy supplied by the battery from the beginning of the test procedure used for certification until the applicable break-off criterion of the test procedure used for certification is reached. DIN DKE Spec 99100 chapter reference: 6.7.2.6		0..1
[SMC] RemainingCapacity	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingCapacity supplementalSemanticId(s): urn:irdi:0173-1#02-ABL840#001 The in-use data attribute on capacity, corresponding with the definition of rated capacity. DIN DKE Spec 99100 chapter reference: 6.7.2.3		0..1

[SML] NegativeEvents	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#negativeEvents The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		0..1
[SML] InformationOnAccidents	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#informationOnAccidents supplementalSemanticId(s): urn:samm:io.admin-shell.idta.shared:3.1.0#DocumentIdentifierSet urn:irdi:0173-1#02-ABL848#001 The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		1
[SMC] TemperatureInformation	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#temperatureInformation The battery passport must include periodically recorded information on the operating environmental conditions, including temperature. DIN DKE Spec 99100 chapter reference: 6.7.7.5 - 8		1
[SMC] RemainingPowerCapability	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingPowerCapability supplementalSemanticId(s): urn:irdi:0173-1#02-ABL820#001 Original power capability (in Watts) and limits, with temperature range when relevant. - The amount of energy that a battery is capable to provide over a given period of time under reference conditions. - Power capability at 80% and 20% state of charge. DIN DKE Spec 99100 chapter reference: 6.7.3.3		0..1
[SMC] EvolutionOfSelfDischarge	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#evolutionOfSelfDischarge Evolution of self-discharge rates is the change of self-discharge over time and usage, as percentage calculated from the initial and current self-discharge rate. DIN DKE Spec 99100 chapter reference: 6.7.4.8		0..1

[SMC] CurrentSelfDischargingRate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#currentSelfDischargingRate The current self-discharge rate is the change of the self-discharge rate in an idle state of the battery in reference conditions (temperature etc.) at aging parameter x, e.g. after a certain amount of storage time or, number of cycles. DIN DKE Spec 99100 chapter reference: 6.7.4.7		0..1
[SMC] RemainingRoundTripEnergyEfficiency	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingRoundTripEnergyEfficiency The battery passport shall include information, where possible, about the remaining round trip energy efficiency as information on the state of health of the battery The update frequency of remaining round trip energy efficiency should be aligned with the update frequency of round trip energy efficiency fade and should be provided upon change of the battery status. DIN DKE Spec 99100 chapter reference: 6.7.4.4		0..1

3.2. Properties of the SMC "EnergyThroughput"

Figure 3 shows the UML-diagram for **SMC EnergyThroughput**.

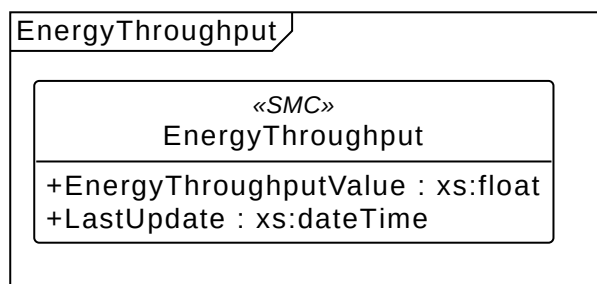


Figure 3. UML-Diagram for SMC "energyThroughput"

Data points related to energy throughput.

idShort:	EnergyThroughput
Class:	SubmodelElementCollection
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#energyThroughput supplementalSemanticId(s): urn:irdi:0173-1#02-ABL850#001
Parent:	ProductCondition

Explanation:	The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.7		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] EnergyThroughputValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#energyThroughputValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL850#001 Overall sum of the energy throughput over the battery lifetime (data attribute should be reported as measured for further potential processing. In addition, the normalisation by usable battery energy could add a further useful value that ensures comparability among battery sizes).	[xs:float] 100000 (kWh)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.3. Properties of the SMC "CapacityThroughput"

Figure 4 shows the UML-diagram for **SMC capacityThroughput**.

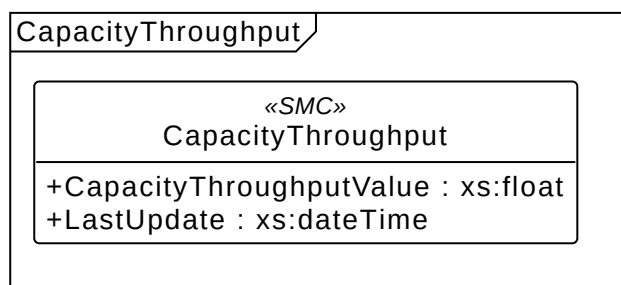


Figure 4. UML-Diagram for SMC "capacityThroughput"

Data points related to capacity throughput.

idShort:	CapacityThroughput		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#capacityThroughput supplementalSemanticId(s): urn:irdi:0173-1#02-ABL849#001		
Parent:	ProductCondition		
Explanation:	The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.8		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] CapacityThroughputValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#capacityThroughputValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL849#001 isCaseOf: 0173-1#02-ABL849#001 Overall sum of the capacity throughput over the battery lifetime (data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity could add a further useful value that ensures comparability among battery sizes).	[xs:float] 214 (Ah)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.4. Properties of the SMC "NumberOfFullCycles"

Figure 5 shows the UML-diagram for **SMC** `numberOfFullCycles`.

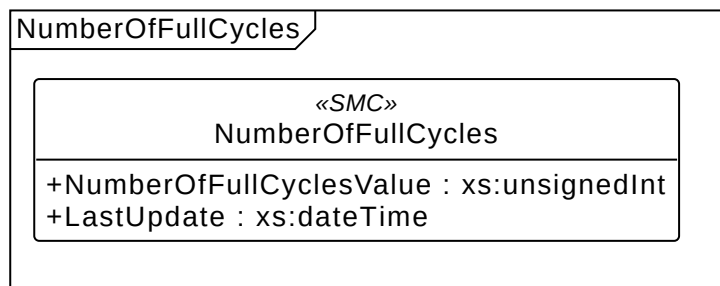


Figure 5. UML-Diagram for SMC "numberOfFullCycles"

Data points related to number of full charging and discharging cycles.

idShort:	NumberOfFullCycles		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#numberOfFullCycles supplementalSemanticId(s): urn:irdi:0173-1#02-ABL868#001		
Parent:	ProductCondition		
Explanation:	Number of (full) charging and discharging cycles. DIN DKE Spec 99100 chapter reference: 6.7.6.4		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] NumberOfFull CyclesValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#numberOfFullCyclesValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL868#001 isCaseOf: 0173-1#02-ABL868#001 Total count of times a battery has been fully charged from empty to full capacity and then fully discharged, used as an indicator of the battery's usage and lifespan.	[xs:unsignedInt] 600 (cycle)	1

[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1
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3.5. Properties of the SMC "StateOfCertifiedEnergy"

Figure 6 shows the UML-diagram for **SMC stateOfCertifiedEnergy**.

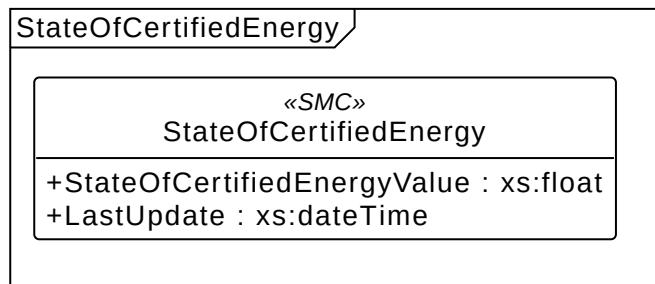


Figure 6. UML-Diagram for SMC "stateOfCertifiedEnergy"

Data points related to the state of certified energy.

idShort:	StateOfCertifiedEnergy		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfCertifiedEnergy supplementalSemanticId(s): urn:irdi:0173-1#02-ABL847#001		
Parent:	ProductCondition		
Explanation:	Definition based on UNECE GTR 22: The measured or on-board UBE performance at a specific point in its lifetime, expressed as a percentage of the certified usable battery energy. DIN DKE Spec 99100 chapter reference: 6.7.2.7		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.

[Prop] StateOfCertifiedEnergyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfCertifiedEnergyValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL847#001 Measured or on-board UBE performance at a specific point in its lifetime, expressed as a percentage of the certified usable battery energy.	[xs:float] 70 (%)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.6. Properties of the SMC "RemainingEnergy"

Figure 7 shows the UML-diagram for **SMC remainingEnergy**.

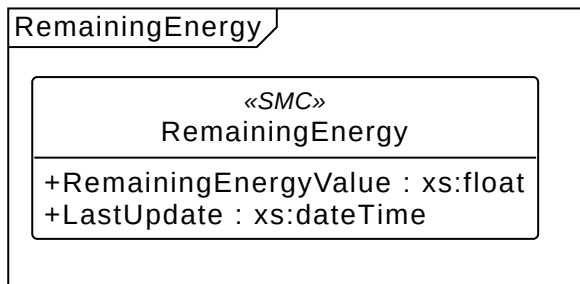


Figure 7. UML-Diagram for SMC "remainingEnergy"

Data points related to remaining energy.

idShort:	RemainingEnergy
Class:	SubmodelElementCollection
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingEnergy supplementalSemanticId(s): urn:irdi:0173-1#02-ABL870#001
Parent:	ProductCondition

Explanation:	Definition from UNECE GTR 22, applicable only to EVs. The energy supplied by the battery from the beginning of the test procedure used for certification until the applicable break-off criterion of the test procedure used for certification is reached. DIN DKE Spec 99100 chapter reference: 6.7.2.6		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] RemainingEnergyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingEnergyValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL870#001 UBE determined at the present point in the lifetime of the vehicle by the test procedure used for certification.	[xs:float] 30 (kWh)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.7. Properties of the SMC "RemainingCapacity"

Figure 8 shows the UML-diagram for **SMC remainingCapacity**.

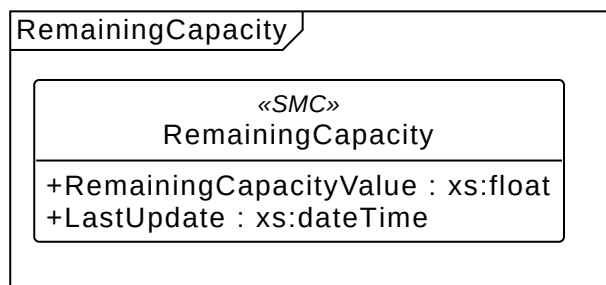


Figure 8. UML-Diagram for SMC "remainingCapacity"

Data points related to remaining capacity.

idShort:	RemainingCapacity
-----------------	--------------------------

Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingCapacity supplementalSemanticId(s): urn:irdi:0173-1#02-ABL840#001		
Parent:	ProductCondition		
Explanation:	The in-use data attribute on capacity, corresponding with the definition of rated capacity. DIN DKE Spec 99100 chapter reference: 6.7.2.3		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] RemainingCapacityValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingCapacityValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL840#001 isCaseOf: 0173-1#02-ABL840#001 In-use data attribute on capacity, corresponding with the definition of rated capacity.	[xs:float] 56 (Ah)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.8. Properties of the SML "NegativeEvents"

UML-Diagram for SML "negativeEvents" shows the UML-diagram for [SML NegativeEvents](#).

UML-Diagram for SML "negativeEvents"

```
Failed to generate image: Permission denied @ rb_sysopen -
/documents/UML_for_SML_negativeEvents.svg
@startuml
!theme plain
```

```

left to right direction
hide class circle
hide class methods
skinparam classAttributeIconSize 0
' skinparam linetype polyline
skinparam linetype ortho

mainframe NegativeEvents

class "NegativeEvents" as ID00000001 <<SML>> {
    +[00] NegativeEvent : SMC [0..*]
}

class "[00] NegativeEvent" as ID00000002 <<SMC>> {
    +NegativeEventValue : xs:string
    +LastUpdate : xs:dateTime
}

ID00000001 *-- ID00000002 : ""
@enduml

```

Data points related to negative events.

idShort:	NegativeEvents		
Class:	SubmodelElementList		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#negativeEvents		
Parent:	ProductCondition		
Explanation:	The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		
Element details:	orderRelevant=No, typeValueListElement=SubmodelElementCollection		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[SMC] NegativeEvent	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#NegativeEvent The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		0..*

3.9. Properties of the SML "InformationOnAccidents"

Figure 9 shows the UML-diagram for **SML InformationOnAccidents**.

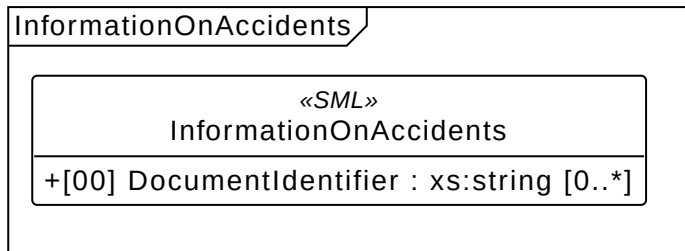


Figure 9. UML-Diagram for SML "InformationOnAccidents"

Data points related to a information on accidents.

idShort:	InformationOnAccidents		
Class:	SubmodelElementList		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#informationOnAccidents supplementalSemanticId(s): urn:samm:io.admin-shell.idta.shared:3.1.0#DocumentIdentifierSet urn:irdi:0173-1#02-ABL848#001		
Parent:	ProductCondition		
Explanation:	The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		
Element details:	orderRelevant=No, typeValueListElement=Property, valueTypeListElement=xs:string		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] DocumentIdentifier	urn:samm:io.admin-shell.idta.handover_documentation:2.0.0#DocumentIdentifier Alphanumeric character sequence uniquely identifying a document.	[xs:string] XF90-884	0..*

3.10. Properties of the SMC "TemperatureInformation"

Figure 10 shows the UML-diagram for **SMC TemperatureInformation**.

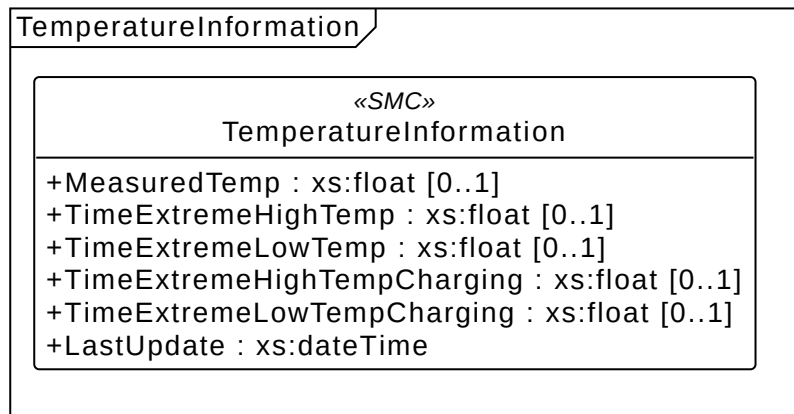


Figure 10. UML-Diagram for SMC "temperatureInformation"

Data points related to temperature.

idShort:	TemperatureInformation		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#temperatureInformation		
Parent:	ProductCondition		
Explanation:	The battery passport must include periodically recorded information on the operating environmental conditions, including temperature. DIN DKE Spec 99100 chapter reference: 6.7.7.5 - 8		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] MeasuredTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#measuredTemp Measured temperature value as part of temperature information.	[xs:float] (degreeCelsius)	0..1
[Prop] TimeExtremeHighTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#timeExtremeHighTemp supplementalSemanticId(s): urn:irdi:0173-1#02-ABL816#001 Cumulated time spent above the given upper boundary of temperature.	[xs:float] 0 (min)	0..1

[Prop] TimeExtremeLowTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#timeExtremeLowTemp supplementalSemanticId(s): urn:irdi:0173-1#02-ABL817#001 Cumulated time spent below the given lower boundary of temperature.	[xs:float] 0 (min)	0..1
[Prop] TimeExtremeHighTempCharging	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#timeExtremeHighTempCharging supplementalSemanticId(s): urn:irdi:0173-1#02-ABL818#001 Cumulated time spent above the given upper boundary of temperature during Charging.	[xs:float] 0 (min)	0..1
[Prop] TimeExtremeLowTempCharging	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#timeExtremeLowTempCharging supplementalSemanticId(s): urn:irdi:0173-1#02-ABL819#001 Cumulated time spent below the given lower boundary of temperature during charging.	[xs:float] 0 (min)	0..1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.11. Properties of the SMC "RemainingPowerCapability"

Figure 11 shows the UML-diagram for **SMC remainingPowerCapability**.

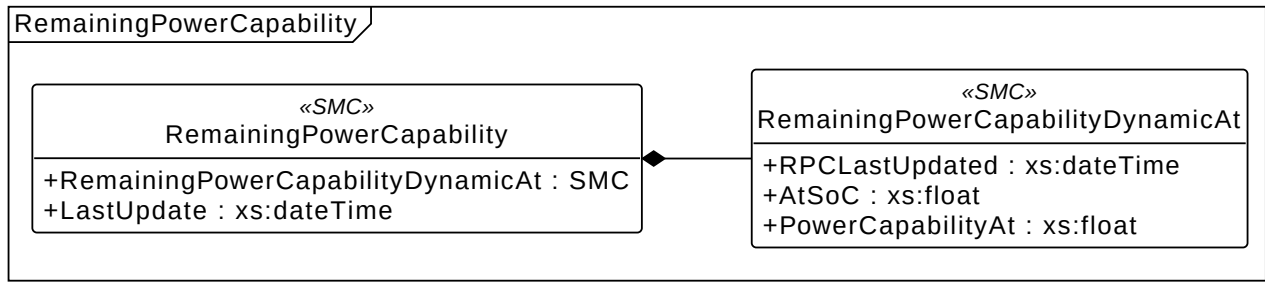


Figure 11. UML-Diagram for SMC "remainingPowerCapability"

Data points related to remaining power capability.

idShort:	RemainingPowerCapability		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingPowerCapability supplementalSemanticId(s): urn:irdi:0173-1#02-ABL820#001		
Parent:	ProductCondition		
Explanation:	Original power capability (in Watts) and limits, with temperature range when relevant. - The amount of energy that a battery is capable to provide over a given period of time under reference conditions. - Power capability at 80% and 20% state of charge. DIN DKE Spec 99100 chapter reference: 6.7.3.3		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[SMC] RemainingPowerCapabilityDynamicAt	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingPowerCapabilityDynamicAt supplementalSemanticId(s): urn:irdi:0173-1#02-ABL853#001 isCaseOf: 0173-1#02-ABL853#001 Defined by 'pre-use power capability (definition of power capability as given in Battery Regulation.		1

[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1
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3.12. Properties of the SMC "EvolutionOfSelfDischarge"

Figure 12 shows the UML-diagram for **SMC evolutionOfSelfDischarge**.

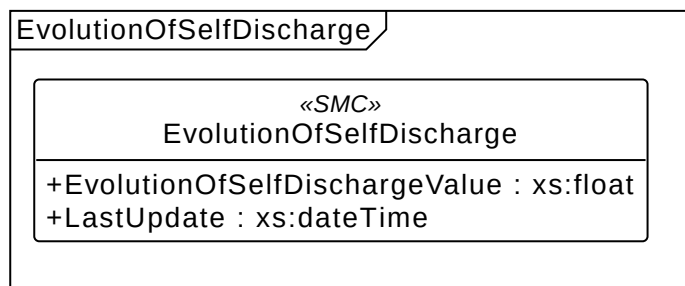


Figure 12. UML-Diagram for SMC "evolutionOfSelfDischarge"

Data points related to evolution of self discharge.

idShort:	EvolutionOfSelfDischarge		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#evolutionOfSelfDischarge		
Parent:	ProductCondition		
Explanation:	Evolution of self-discharge rates is the change of self-discharge over time and usage, as percentage calculated from the initial and current self-discharge rate. DIN DKE Spec 99100 chapter reference: 6.7.4.8		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.

[Prop] EvolutionOfSelfDischargeValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#evolutionOfSelfDischargeValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL834#001 isCaseOf: 0173-1#02-ABL834#001 Total count of times a battery has been fully charged from empty to full capacity and then fully discharged, used as an indicator of the battery's usage and lifespan.	[xs:float] 1 (%)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.13. Properties of the SMC "CurrentSelfDischargingRate"

Figure 13 shows the UML-diagram for **SMC currentSelfDischargingRate**.

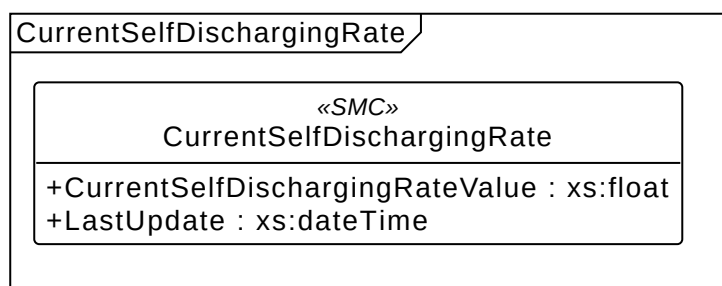


Figure 13. UML-Diagram for SMC "currentSelfDischargingRate"

Data points related to current self discharging rate.

idShort:	CurrentSelfDischargingRate
Class:	SubmodelElementCollection
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#currentSelfDischargingRate
Parent:	ProductCondition

Explanation:	The current self-discharge rate is the change of the self-discharge rate in an idle state of the battery in reference conditions (temperature etc.) at aging parameter x, e.g. after a certain amount of storage time or, number of cycles. DIN DKE Spec 99100 chapter reference: 6.7.4.7		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] CurrentSelfDischargingRateValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#currentSelfDischargingRateValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL825#001 isCaseOf: 0173-1#02-ABL825#001 Current self-discharge in % of capacity per unit of time in defined conditions (temperature range) during the use phase.	[xs:float] 2 (%)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.14. Properties of the SMC

"RemainingRoundTripEnergyEfficiency"

Figure 14 shows the UML-diagram for **SMC remainingRoundTripEnergyEfficiency**.

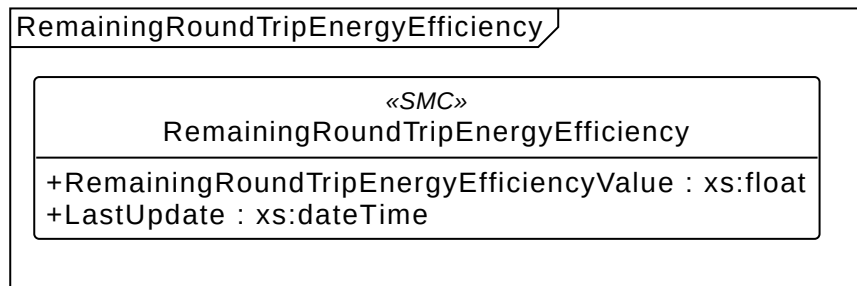


Figure 14. UML-Diagram for SMC "currentSelfDischargingRate"

Data points related to remaining round trip energy efficiency.

idShort:	RemainingRoundTripEnergyEfficiency		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingRoundTripEnergyEfficiency		
Parent:	ProductCondition		
Explanation:	The battery passport shall include information, where possible, about the remaining round trip energy efficiency as information on the state of health of the battery The update frequency of remaining round trip energy efficiency should be aligned with the update frequency of round trip energy efficiency fade and should be provided upon change of the battery status. DIN DKE Spec 99100 chapter reference: 6.7.4.4		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] RemainingRoundTripEnergyEfficiencyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingRoundTripEnergyEfficiencyValue supplementalSemanticId(s): urn:iridi:0173-1#02-ABL851#001 isCaseOf: 0173-1#02-ABL851#001 Percentage of energy that can be recovered and used after a full charge and discharge cycle, reflecting the battery's ability to retain and return energy efficiently over time.	[xs:float] 80 (%)	1

[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1
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3.15. Properties of the SMC "StateOfCharge"

Figure 15 shows the UML-diagram for **SMC stateOfCharge**.

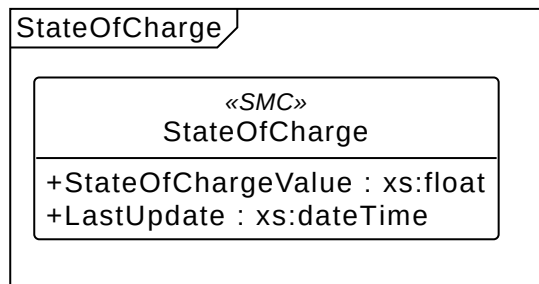


Figure 15. UML-Diagram for SMC "stateOfCharge"

Data points related to state of charge.

idShort:	StateOfCharge		
Class:	SubmodelElementCollection		
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfCharge isCaseOf: urn:irdi:0173-1#02-ABL821#002		
Parent:	ProductCondition		
Explanation:	Available capacity in a battery expressed as a percentage of remaining capacity.		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.

[Prop] StateOfCharge Value	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#stateOfChargeValue supplementalSemanticId(s): urn:irdi:0173-1#02-ABL821#001 isCaseOf: 0173-1#02-ABL821#002 Available capacity in a battery expressed as a percentage of remaining capacity.	[xs:float] 70 (%)	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.16. Properties of the SMC "NegativeEvent"

Figure 16 shows the UML-diagram for **SMC NegativeEvent**.

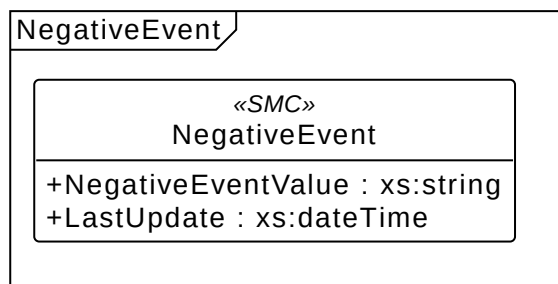


Figure 16. UML-Diagram for SMC "NegativeEvent"

Data points related to a negative event.

idShort:	NegativeEvent
Class:	SubmodelElementCollection
Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#NegativeEvent
Parent:	NegativeEvents

Explanation:	The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] NegativeEvent Value	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#negativeEventValue Negative event, such as an accident. No further definition provided by regulation.	[xs:string] overcharged	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#lastUpdate supplementalSemanticId(s): urn:irdi:0173-1#02-ABG740#003 isCaseOf: 0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[xs:dateTime] 2000-01-01T14:23:00	1

3.17. Properties of the SMC

"RemainingPowerCapabilityDynamicAt"

Figure 17 shows the UML-diagram for **SMC RemainingPowerCapabilityDynamicAt**.

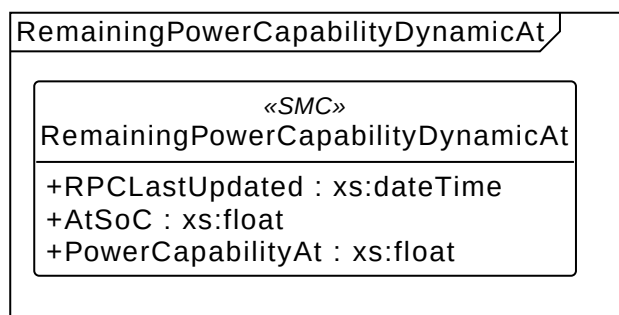


Figure 17. UML-Diagram for SMC "RemainingPowerCapabilityDynamicAt"

Data points of remaining power capability.

idShort:	RemainingPowerCapabilityDynamicAt
Class:	SubmodelElementCollection

Semantic id(s):	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#remainingPowerCapabilityDynamicAt supplementalSemanticId(s): urn:irdi:0173-1#02-ABL853#001 isCaseOf: 0173-1#02-ABL853#001		
Parent:	RemainingPowerCapability		
Explanation:	Defined by 'pre-use power capability (definition of power capability as given in Battery Regulation.		
Element details:	-		
[SME type] idShort	semantic id(s) Definition@en Description@en	[valueType] example (unit)	card.
[Prop] RPCLastUpdated	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#rPCLastUpdated RPC last updated.	[xs:dateTime] 2000-01-01T14:23:00	1
[Prop] AtSoC	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#atSoC supplementalSemanticId(s): urn:irdi:0173-1#02-ABL821#001 isCaseOf: 0173-1#02-ABL821#001 Available capacity in a battery expressed as a percentage of remaining capacity.	[xs:float] 80 (%)	1
[Prop] PowerCapabilityAt	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2#powerCapabilityAt Power capability at.	[xs:float] 95 (%)	1

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] from the second information.
- The types of SubmodelElements are abbreviated (see [Table 1](#)):

Table 1. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement
SML	SubmodelElementList

- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

Annex B. Changes to the submodel template

General

This annex lists the changes from version to version of the Submodel, together with major changes in the overall document.

Changes Version 1.0.2

Major changes:

- fix measurement unit of property "remainingCapacityValue" from kilowattHour to AmpereHour (Ah)
- increase version of semanticIds and ConceptDescription/ids to urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.2
- in aasx: add/correct unit, symbol and unitId to ConceptDescriptions if missing
- use new version of documentation generator (measurement units now displayed)

Known deviation to LongList BatteryPassport V1.3.1

- data type of capacity with unit Ah is xsd:float and not xsd:integer

Changes Version 1.0.1

Major changes:

- new optional property "measuredTemp" as part of TemperatureInformation (<https://github.com/admin-shell-io/smt-semantic-models/issues/80>)

Changes Version 1.0

- First Version conformant to DIN DKE SPEC 99100

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